THE PROBLEM

Earth is 40 minutes away

Earth-based symptom checkers assume Earth. In deep space, a stuffy nose on day 2 is fluid shift, not a cold — and joint pain after a spacewalk is decompression sickness until proven otherwise.

> **When the doctor is 20 minutes away, the crew still needs an answer now.**

With communication blackouts of up to two weeks, a crew with no physician aboard has to decide what to do. VITALS answers three questions:

What could this be?
Conditions from a research-grounded knowledge base — no internet.

How serious is it?
Urgency first, so the crew knows if a ground call is needed.

What do we do now?
Immediate steps and a safety backstop for non-medical crewmembers.

TEAM TETHER

Who built it

Joshua Collado
Project lead & main coder

Joaquin
Medical research & co-coder

Cruz
Interface design & user testing

Ms. Sylvia Odoms
Science Academy teacher

South Texas ISD Science Academy · Mercedes, Texas

NASA HUNCH 2026–27 — Software & Technology
SFT: Medical Diagnostic System with Machine Learning & Artificial Intelligence

NEXT, FOR CDR

Ground crew & doctor screens · Jetson Orin Nano server · first sensor (pulse-ox) · user testing with 5+ people

Scan for the site
joshuacollado636
.wixsite.com/vitals

VITALS is a student prototype for diagnosis and decision support — a doctor has the final say. Not a medical device.

VITALS

An onboard medical assistant for astronauts on long-duration Moon and Mars missions.

TEAM TETHER
NASA HUNCH 2026–27

THE VITALS CORE

Built to be trusted

Talk in plain words
Describe symptoms the way you'd say them out loud.

Urgency first
Every answer leads with how serious it is. Critical conditions are flagged at once.

Source-backed answers
Grounded in NASA research. It never invents a condition — it only names what it knows.

Safety backstop
A rules engine can always raise the alarm, even if the AI doesn't.

Fix what it misread
The *Wrong?* button lets the crew correct it. Every case is logged.

Fully offline
A local AI model and a cited knowledge base. No internet needed.

> **A doctor has the final say. VITALS helps the crew get there.**

HOW IT WORKS

How VITALS Works

01 Ask
Describe symptoms in plain English.

02 Scope
Medical question, or a plain reply?

03 Read
Pull out the symptoms named.

04 Retrieve
Search 35 cited spaceflight conditions.

05 Reason
Local AI reads only the cited pages.

06 Check
Rules engine blocks anything invented.

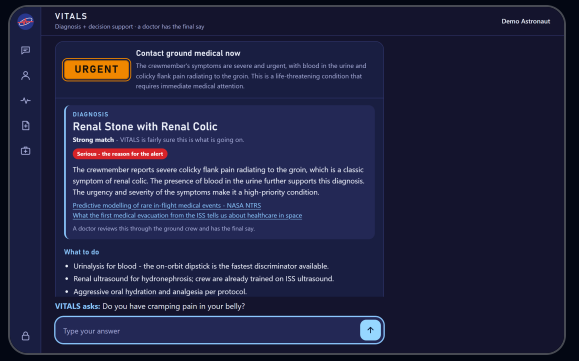
07 Answer
Urgency, what it could be, and what to do now.

No in-flight data to train on — so VITALS grounds a local AI in cited NASA research instead.

35 SPACEFLIGHT CONDITIONS	**262** AUTOMATED TESTS PASSED
18/19 MUST-ESCALATE CAUGHT	**6/6** BASIC MISSIONS PASSED

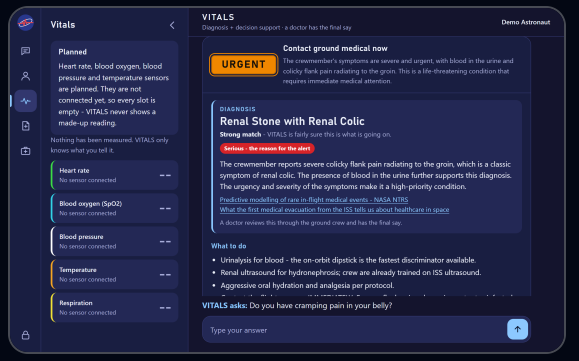
THE PROTOTYPE

Working now



The astronaut screen: one complaint in, urgency and next steps out — with the research it came from.

Astronaut screen ✓ · Offline AI ✓ · Rules backstop ✓ · Ground crew screen · planned · Doctor review · planned · Sensors · planned



Vitals, Doctor notes and Supplies tabs are in the app now, marked *planned*.

AI MODEL **Llama 3.2 (3B)**	HARDWARE **Jetson Orin Nano**
SAFETY **Deterministic rules**	KNOWLEDGE **35 NASA conditions**